

Exp No: 25

PRODUCER-CONSUMER PROBLEM USING SYSTEMC

Program

```
#include <systemc.h>

// Producer Module
SC_MODULE(Producer) {
    sc_fifo_out<int> out;
    void produce() {
        for (int i = 1; i <= 5; i++) {
            cout << "Produced: " << i << endl;
            out.write(i);
            wait(1, SC_SEC);
        }
    }
    SC_CTOR(Producer) {
        SC_THREAD(produce);
    }
};

// Consumer Module
SC_MODULE(Consumer) {
    sc_fifo_in<int> in;
    void consume() {
        int value;
        while (true) {
            in.read(value);
            cout << "Consumed: " << value << endl;
            wait(1, SC_SEC);
        }
    }
    SC_CTOR(Consumer) {
        SC_THREAD(consume);
    }
};
```

```
};  
  
int sc_main(int argc, char* argv[]) {  
    sc_fifo<int> fifo(5);  
    Producer producer("Producer");  
    Consumer consumer("Consumer");  
    producer.out(fifo);  
    consumer.in(fifo);  
    sc_start(10, SC_SEC);  
    return 0;  
}
```

Output

```
SystemC 2.3.3-Accellera --- May 23 2020 14:33:56  
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Produced: 1  
Consumed: 1  
Produced: 2  
Consumed: 2  
Produced: 3  
Consumed: 3  
Produced: 4  
Consumed: 4  
Produced: 5  
Consumed: 5  
Done
```